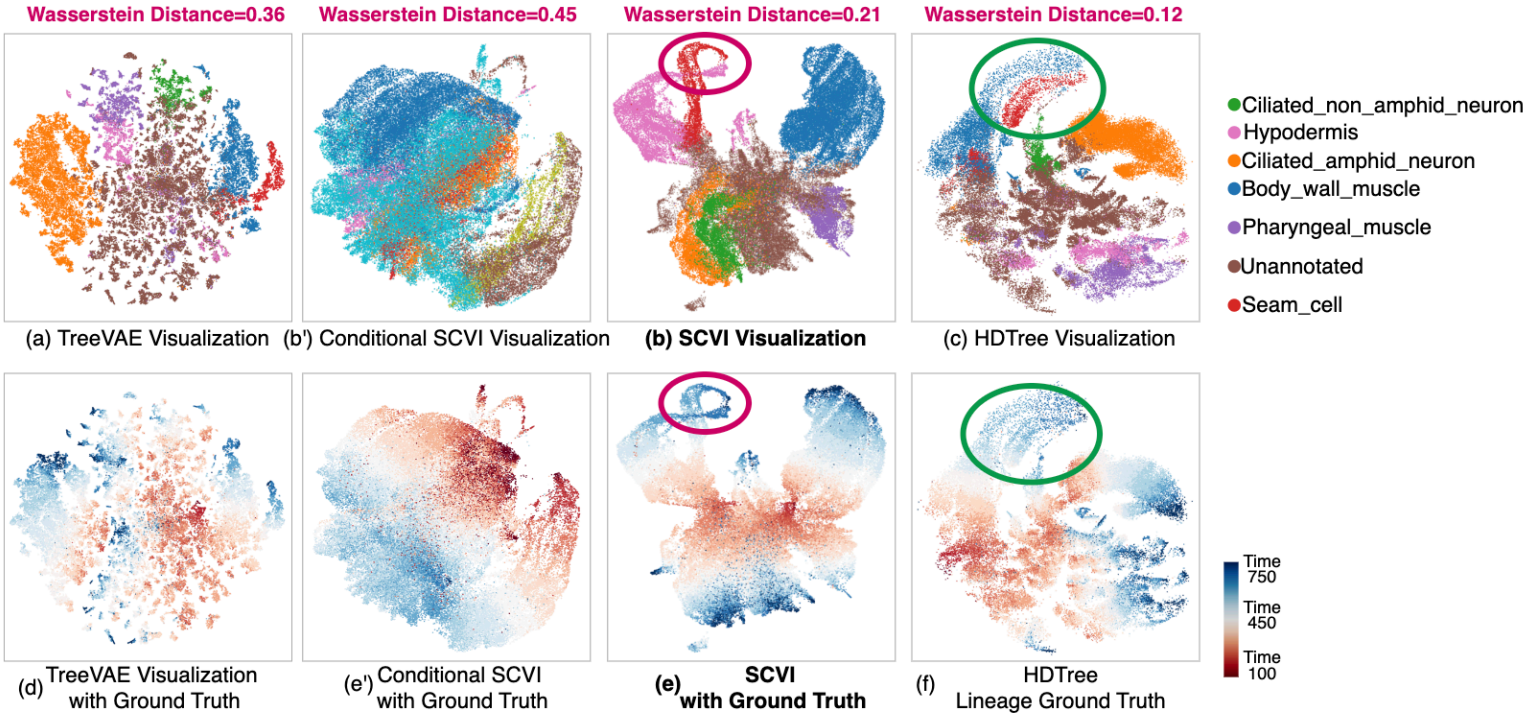

Figure R4.1: Latent space comparison on C. elegans dataset. To address the concern regarding the necessity of HDTree compared to simpler models, we visualize the latent representations colored by cell types (top row) and developmental time (bottom row). (b, e) Conditional scVI maps cells into a relatively flat, monolithic manifold. While it aligns with time gradients, it fails to separate distinct developmental bifurcations. (c, f) HDTree, driven by its hierarchical inductive bias (LHQL), explicitly uncovers a highly structured, multi-branching topology. It visually traces the continuous biological progression from early progenitor states (red, unannotated center) to distinct, mature terminal cell types (blue, outward branches), corroborating its superior capability for de novo lineage tracing without relying on perfect labels.(Note: Red circles highlight regions in scVI where distinct temporal stages or distinct lineages physically entangle due to manifold overlap. Green circles show how HDTree resolves this, maintaining strict temporal progression and clean hierarchical separation without short-circuiting.)

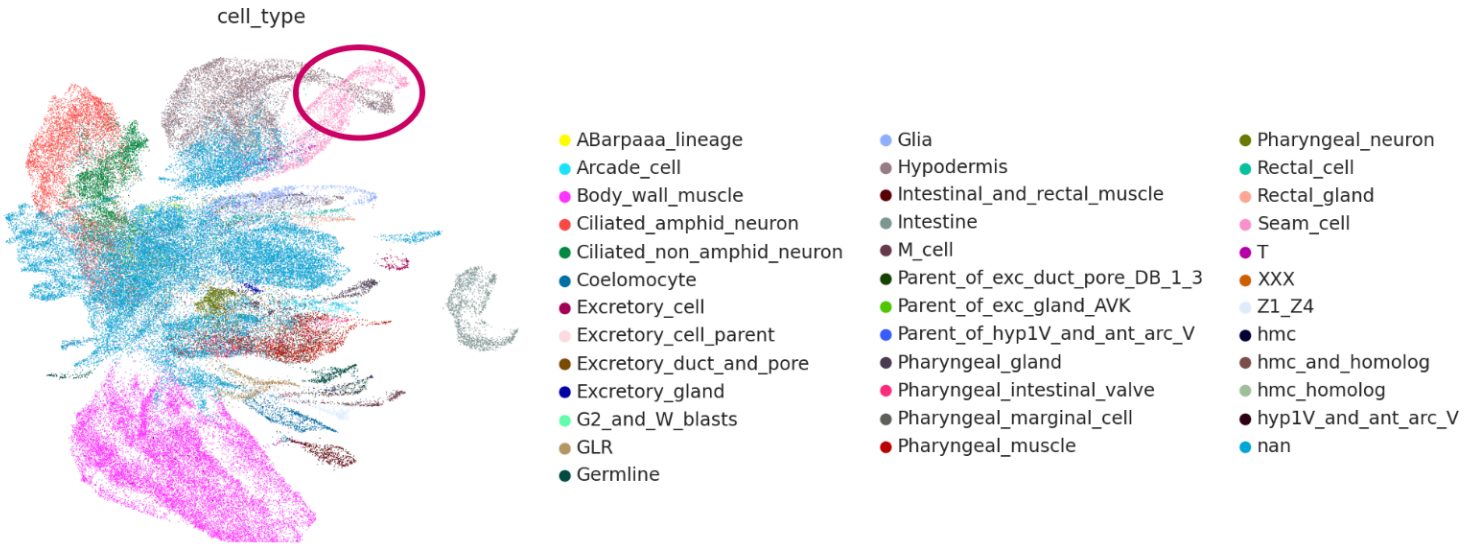

Figure R4.1': The results provided by Reviewer.https://docs.scvi-tools.org/en/stable/tutorials/notebooks/scrna/scVI_DE_worm.html#